```
> restart: unprotect('D'):

  m := 8;

  d := 12;                    degre := d + 1:

  twist := - 7;

  ##

  #SS := x^(d+1) + 1:
  SS := x^(d+1) + x^6 + 1:
  #SS := x^(d+1) + y^2*x^3 + x^4 + x^2 + 1:

  ##

  Jet_xR := expand(

  + (1/Ry^(m-0*2)) * add(AA||j * x1^(m-0*3-j) * (R1/RR)^j *
  Delta_xR^0, j=0..m-0*3)

  + (1/Ry^(m-1*2)) * add(BB||j * x1^(m-1*3-j) * (R1/RR)^j *
  Delta_xR^1, j=0..m-1*3)

  + (1/Ry^(m-2*2)) * add(CC||j * x1^(m-2*3-j) * (R1/RR)^j *
  Delta_xR^2, j=0..m-2*3)

  + (1/Ry^(m-3*2)) * add(DD||j * x1^(m-3*3-j) * (R1/RR)^j *
  Delta_xR^3, j=0..m-3*3)

  + (1/Ry^(m-4*2)) * add(EE||j * x1^(m-4*3-j) * (R1/RR)^j *
  Delta_xR^4, j=0..m-4*3)

  + (1/Ry^(m-5*2)) * add(FF||j * x1^(m-5*3-j) * (R1/RR)^j *
  Delta_xR^5, j=0..m-5*3)

  + (1/Ry^(m-6*2)) * add(GG||j * x1^(m-6*3-j) * (R1/RR)^j *
  Delta_xR^6, j=0..m-6*3)

  + (1/Ry^(m-7*2)) * add(HH||j * x1^(m-7*3-j) * (R1/RR)^j *
  Delta_xR^7, j=0..m-7*3) ):

  ##

  for l from 0 to m - 0*3 do dA[l] := (m-0*2)*d - (m-0*3-l)*2 - l*1
  - 0*4 + twist: od:

  for l from 0 to m - 1*3 do dB[l] := (m-1*2)*d - (m-1*3-l)*2 - l*1
  - 1*4 + twist: od:

  for l from 0 to m - 2*3 do dC[l] := (m-2*2)*d - (m-2*3-l)*2 - l*1
  - 2*4 + twist: od:
```

```
for l from 0 to m - 3*3 do dD[l] := (m-3*2)*d - (m-3*3-l)*2 - l*1
- 3*4 + twist: od:

for l from 0 to m - 4*3 do dE[l] := (m-4*2)*d - (m-4*3-l)*2 - l*1
- 4*4 + twist: od:

for l from 0 to m - 5*3 do dF[l] := (m-5*2)*d - (m-5*3-l)*2 - l*1
- 5*4 + twist: od:

for l from 0 to m - 6*3 do dG[l] := (m-6*2)*d - (m-6*3-l)*2 - l*1
- 6*4 + twist: od:

for l from 0 to m - 7*3 do dH[l] := (m-7*2)*d - (m-7*3-l)*2 - l*1
- 7*4 + twist: od:

##

Delta_xR := x1 * (R2/RR - R1*R1/RR^2) - x2 * (R1/RR):

Solx1 := - y1*Ry/Rx + R1/Rx:

Solx2 := - Rxx*y1^2*Ry^2/Rx^3 + 2*Rxx*y1*Ry*R1/Rx^3 - Rxx*
R1^2/Rx^3 + 2*y1^2*Rxy*Ry/Rx^2
         - 2*y1*Rxy*R1/Rx^2 - y1^2*Ryy/Rx - y2*Ry/Rx + R2/Rx:

##

Jet_Ry := sort(mtaylor(

                        expand(subs(R2=-(RR^2*Delta_Ry-RR*R1*y2-
R1^2*y1)/(RR*y1),
                        expand(subs({x1=Solx1,x2=Solx2},
                 Jet_xR)))),

         [R1,y1,Delta_Ry], 100), [R1,y1,Delta_Ry]):

##

printlevel := 2:
for k from 0 to m/3   do
for j from 0 to m-3*k do

EEq[m-j-3*k,j,k] := factor(

  Ry^(m-j-3*k)*RR^(m-j-3*k)*Rx^(m-2*k)
*
  coeftayl(Jet_Ry, [R1,y1,Delta_Ry]=[0,0,0], [m-j-3*k,j,k]) ):

od: od:

## EQUATION RR EXPLICITE

RR := y^(d+1) + SS;

Rx := diff(RR,x):
Ry := diff(RR,y):
```

```
Rxx := diff(RR,x,x):
Rxy := diff(RR,x,y):
Ryy := diff(RR,y,y):

##

printlevel := 2:
for k from 0 to m/3   do
for j from 0 to m-3*k do

Eq[m-j-3*k,j,k] := mtaylor(EEq[m-j-3*k,j,k], y, (m-j-3*k)*d):

Var[m-j-3*k,j,k] := indets(Eq[m-j-3*k,j,k]) minus {x,y}:

od: od:

##
```

$$m := 8$$

$$d := 12$$

$$twist := -7$$

$$RR := x^{13} + y^{13} + x^{6} + 1 \quad \textbf{(1)}$$

```
> ## EXPAND ALL UNKNOWNS UP TO y^(1*d)

for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
[l],1*d-1)): od:

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
[l],1*d-1)): od:

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
[l],1*d-1)): od:

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
[l],1*d-1)): od:

for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
[l],1*d-1)): od;

for l from 0 to m - 5*3 do
FF||l := add(add(F||l[i,j]*x^i*y^j, i=0..dF[l]-j), j=0..min(dF
[l],1*d-1)): od:

for l from 0 to m - 6*3 do:
GG||l := add(add(G||l[i,j]*x^i*y^j, i=0..dG[l]-j), j=0..min(dG
```

```
[l],1*d-1)): od:

for l from 0 to m - 7*3 do
HH||l := add(add(H||l[i,j]*x^i*y^j, i=0..dH[l]-j), j=0..min(dH
[l],1*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(1*d)

Systemey1d := {}:

for k from 0 to (m-1)/3   do

Coeffs_y1d[1,m-1-3*k,k] := {coeffs(expand(mtaylor(Eq[1,m-1-3*k,
k], y, 1*d)), [x,y])}:

Systemey1d := Systemey1d union Coeffs_y1d[1,m-1-3*k,k]:

od:

Systeme_y1d := Systemey1d:

## VARIABLES FOR DIVISIBILITY BY y^(1*d)

Variables_y1d := indets(Systeme_y1d):

## RESOLUTION y^(1*d)

Solutions_y1d := solve(Systeme_y1d, Variables_y1d):

assign(Solutions_y1d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;
for l from 0 to m - 5*3 do nops(expand(FF||l)) od;
for l from 0 to m - 6*3 do nops(expand(GG||l)) od;
for l from 0 to m - 7*3 do nops(expand(HH||l)) od;
```

$$l := 0$$

$$l := 0$$

$$4755$$

$$l := 1$$

$$11939$$

$$l := 2$$

$$846$$

$$l := 3$$

858

$l := 4$

870

$l := 5$

882

$l := 6$

894

$l := 7$

906

$l := 8$

918

$l := 9$

$l := 0$

3983

$l := 1$

7701

$l := 2$

582

$l := 3$

594

$l := 4$

606

$l := 5$

618

$l := 6$

$l := 0$

586

$l := 1$

843

$l := 2$

318

$l := 3$

$l := 0$

$l := 0$

$l := 0$

$l := 0$

$l := 0$ **(2)**

```
> ## EXPAND ALL UNKNOWNS UP TO y^(2*d)

  for l from 0 to m - 0*3 do
  AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
  [l],2*d-1)): od:

  for l from 0 to m - 1*3 do
  BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
  [l],2*d-1)): od:

  for l from 0 to m - 2*3 do
  CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
  [l],2*d-1)): od:

  for l from 0 to m - 3*3 do
  DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
  [l],2*d-1)): od:

  for l from 0 to m - 4*3 do
  EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
  [l],2*d-1)): od;

  for l from 0 to m - 5*3 do
  FF||l := add(add(F||l[i,j]*x^i*y^j, i=0..dF[l]-j), j=0..min(dF
  [l],2*d-1)): od:

  for l from 0 to m - 6*3 do:
  GG||l := add(add(G||l[i,j]*x^i*y^j, i=0..dG[l]-j), j=0..min(dG
  [l],2*d-1)): od:

  for l from 0 to m - 7*3 do
  HH||l := add(add(H||l[i,j]*x^i*y^j, i=0..dH[l]-j), j=0..min(dH
  [l],2*d-1)): od:

  ## SYSTEM FOR DIVISIBILITY BY y^(2*d)

  Systemey2d := {}:

  for k from 0 to (m-2)/3   do

  Coeffs_y2d[2,m-2-3*k,k] := {coeffs(expand(mtaylor(Eq[2,m-2-3*k,
  k], y, 2*d)), [x,y])}:

  Systemey2d := Systemey2d union Coeffs_y2d[2,m-2-3*k,k]:

  od:

  Systeme_y2d := Systemey2d:

  ## VARIABLES FOR DIVISIBILITY BY y^(2*d)

  Variables_y2d := indets(Systeme_y2d):

  ## RESOLUTION y^(2*d)
```

```
Solutions_y2d := solve(Systeme_y2d, Variables_y2d):

assign(Solutions_y2d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;
for l from 0 to m - 5*3 do nops(expand(FF||l)) od;
for l from 0 to m - 6*3 do nops(expand(GG||l)) od;
for l from 0 to m - 7*3 do nops(expand(HH||l)) od;
```

$$l := 0$$

$$l := 0$$

$$10769$$

$$l := 1$$

$$20009$$

$$l := 2$$

$$36473$$

$$l := 3$$

$$1572$$

$$l := 4$$

$$1596$$

$$l := 5$$

$$1620$$

$$l := 6$$

$$1644$$

$$l := 7$$

$$1668$$

$$l := 8$$

$$1692$$

$$l := 9$$

$$l := 0$$

$$4482$$

$$l := 1$$

$$8431$$

$$l := 2$$

$$15254$$

$$l := 3$$

$$1044$$

$$l := 4$$
$$1068$$
$$l := 5$$
$$1092$$
$$l := 6$$
$$l := 0$$
$$253$$
$$l := 1$$
$$424$$
$$l := 2$$
$$546$$
$$l := 3$$
$$l := 0$$
$$l := 0$$
$$l := 0$$
$$l := 0$$
$$l := 0 \tag{3}$$

```
> ## EXPAND ALL UNKNOWNS UP TO y^(3*d)

  for l from 0 to m - 0*3 do
  AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
  [l],3*d-1)): od:

  for l from 0 to m - 1*3 do
  BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
  [l],3*d-1)): od:

  for l from 0 to m - 2*3 do
  CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
  [l],3*d-1)): od:

  for l from 0 to m - 3*3 do
  DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
  [l],3*d-1)): od:

  for l from 0 to m - 4*3 do
  EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
  [l],3*d-1)): od;

  for l from 0 to m - 5*3 do
  FF||l := add(add(F||l[i,j]*x^i*y^j, i=0..dF[l]-j), j=0..min(dF
  [l],3*d-1)): od:

  for l from 0 to m - 6*3 do:
```

```
GG||l := add(add(G||l[i,j]*x^i*y^j, i=0..dG[l]-j), j=0..min(dG
[l],3*d-1)): od:

for l from 0 to m - 7*3 do
HH||l := add(add(H||l[i,j]*x^i*y^j, i=0..dH[l]-j), j=0..min(dH
[l],3*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(3*d)

Systemey3d := {}:

for k from 0 to (m-3)/3   do

Coeffs_y3d[3,m-3-3*k,k] := {coeffs(expand(mtaylor(Eq[3,m-3-3*k,
k], y, 3*d)), [x,y])}:

Systemey3d := Systemey3d union Coeffs_y3d[3,m-3-3*k,k]:

od:

Systeme_y3d := Systemey3d:

## VARIABLES FOR DIVISIBILITY BY y^(3*d)

Variables_y3d := indets(Systeme_y3d):

## RESOLUTION y^(3*d)

Solutions_y3d := solve(Systeme_y3d, Variables_y3d):

assign(Solutions_y3d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;
for l from 0 to m - 5*3 do nops(expand(FF||l)) od;
for l from 0 to m - 6*3 do nops(expand(GG||l)) od;
for l from 0 to m - 7*3 do nops(expand(HH||l)) od;
```

$$l := 0$$

$$l := 0$$

$$23130$$

$$l := 1$$

$$27116$$

$$l := 2$$

$$40320$$

$$l := 3$$

$$65375$$
$$l := 4$$
$$2178$$
$$l := 5$$
$$2214$$
$$l := 6$$
$$2250$$
$$l := 7$$
$$2286$$
$$l := 8$$
$$2322$$
$$l := 9$$
$$l := 0$$
$$6157$$
$$l := 1$$
$$8427$$
$$l := 2$$
$$13515$$
$$l := 3$$
$$18497$$
$$l := 4$$
$$1386$$
$$l := 5$$
$$1422$$
$$l := 6$$
$$l := 0$$
$$993$$
$$l := 1$$
$$1915$$
$$l := 2$$
$$2391$$
$$l := 3$$
$$l := 0$$
$$l := 0$$
$$l := 0$$
$$l := 0$$
$$l := 0 \tag{4}$$

```
> ## EXPAND ALL UNKNOWNS UP TO y^(4*d)

  for l from 0 to m - 0*3 do
  AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
  [l],4*d-1)): od:

  for l from 0 to m - 1*3 do
  BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
  [l],4*d-1)): od:

  for l from 0 to m - 2*3 do
  CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
  [l],4*d-1)): od:

  for l from 0 to m - 3*3 do
  DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
  [l],4*d-1)): od:

  for l from 0 to m - 4*3 do
  EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
  [l],4*d-1)): od;

  for l from 0 to m - 5*3 do
  FF||l := add(add(F||l[i,j]*x^i*y^j, i=0..dF[l]-j), j=0..min(dF
  [l],4*d-1)): od:

  for l from 0 to m - 6*3 do:
  GG||l := add(add(G||l[i,j]*x^i*y^j, i=0..dG[l]-j), j=0..min(dG
  [l],4*d-1)): od:

  for l from 0 to m - 7*3 do
  HH||l := add(add(H||l[i,j]*x^i*y^j, i=0..dH[l]-j), j=0..min(dH
  [l],4*d-1)): od:

  ## SYSTEM FOR DIVISIBILITY BY y^(4*d)

  Systemey4d := {}:

  for k from 0 to (m-4)/3   do

  Coeffs_y4d[4,m-4-3*k,k] := {coeffs(expand(mtaylor(Eq[4,m-4-3*k,
  k], y, 4*d)), [x,y])}:

  Systemey4d := Systemey4d union Coeffs_y4d[4,m-4-3*k,k]:

  od:

  Systeme_y4d := Systemey4d:

  ## VARIABLES FOR DIVISIBILITY BY y^(4*d)

  Variables_y4d := indets(Systeme_y4d):

  ## RESOLUTION y^(4*d)
```

```
Solutions_y4d := solve(Systeme_y4d, Variables_y4d):

assign(Solutions_y4d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;
for l from 0 to m - 5*3 do nops(expand(FF||l)) od;
for l from 0 to m - 6*3 do nops(expand(GG||l)) od;
for l from 0 to m - 7*3 do nops(expand(HH||l)) od;
```

$$l := 0$$

$$l := 0$$

$$34830$$

$$l := 1$$

$$38206$$

$$l := 2$$

$$40857$$

$$l := 3$$

$$52887$$

$$l := 4$$

$$49000$$

$$l := 5$$

$$2664$$

$$l := 6$$

$$2712$$

$$l := 7$$

$$2760$$

$$l := 8$$

$$2808$$

$$l := 9$$

$$l := 0$$

$$487$$

$$l := 1$$

$$1073$$

$$l := 2$$

$$1537$$

$$l := 3$$

$$2287$$

$$l := 4$$
$$2184$$
$$l := 5$$
$$1608$$
$$l := 6$$
$$l := 0$$
$$352$$
$$l := 1$$
$$569$$
$$l := 2$$
$$861$$
$$l := 3$$
$$l := 0$$
$$l := 0$$
$$l := 0$$
$$l := 0$$
$$l := 0 \tag{5}$$

```
> ## EXPAND ALL UNKNOWNS UP TO y^(5*d)

  for l from 0 to m - 0*3 do
  AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
  [l],5*d-1)): od:

  for l from 0 to m - 1*3 do
  BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
  [l],5*d-1)): od:

  for l from 0 to m - 2*3 do
  CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
  [l],5*d-1)): od:

  for l from 0 to m - 3*3 do
  DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
  [l],5*d-1)): od:

  for l from 0 to m - 4*3 do
  EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
  [l],5*d-1)): od;

  for l from 0 to m - 5*3 do
  FF||l := add(add(F||l[i,j]*x^i*y^j, i=0..dF[l]-j), j=0..min(dF
  [l],5*d-1)): od:

  for l from 0 to m - 6*3 do:
```

```
GG||l := add(add(G||l[i,j]*x^i*y^j, i=0..dG[l]-j), j=0..min(dG
[l],5*d-1)): od:

for l from 0 to m - 7*3 do
HH||l := add(add(H||l[i,j]*x^i*y^j, i=0..dH[l]-j), j=0..min(dH
[l],5*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(5*d)

Systemey5d := {}:

for k from 0 to (m-5)/3   do

Coeffs_y5d[5,m-5-3*k,k] := {coeffs(expand(mtaylor(Eq[5,m-5-3*k,
k], y, 5*d)), [x,y])}:

Systemey5d := Systemey5d union Coeffs_y5d[5,m-5-3*k,k]:

od:

Systeme_y5d := Systemey5d:

## VARIABLES FOR DIVISIBILITY BY y^(5*d)

Variables_y5d := indets(Systeme_y5d):

## RESOLUTION y^(5*d)

Solutions_y5d := solve(Systeme_y5d, Variables_y5d):

assign(Solutions_y5d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;
for l from 0 to m - 5*3 do nops(expand(FF||l)) od;
for l from 0 to m - 6*3 do nops(expand(GG||l)) od;
for l from 0 to m - 7*3 do nops(expand(HH||l)) od;
```

$$l := 0$$

$$l := 0$$

$$15013$$

$$l := 1$$

$$19501$$

$$l := 2$$

$$21213$$

$$l := 3$$

$$
\begin{aligned}
&25119\\
&l := 4\\
&27095\\
&l := 5\\
&27708\\
&l := 6\\
&3030\\
&l := 7\\
&3090\\
&l := 8\\
&3150\\
&l := 9\\
&l := 0\\
&1\\
&l := 1\\
&1\\
&l := 2\\
&1\\
&l := 3\\
&1\\
&l := 4\\
&1\\
&l := 5\\
&1\\
&l := 6\\
&l := 0\\
&1\\
&l := 1\\
&1\\
&l := 2\\
&1\\
&l := 3\\
&l := 0\\
&l := 0\\
&l := 0\\
&l := 0\\
&l := 0
\end{aligned}
\tag{6}
$$

```
> ## EXPAND ALL UNKNOWNS UP TO y^(6*d)

  for l from 0 to m - 0*3 do
  AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
  [l],6*d-1)): od:

  for l from 0 to m - 1*3 do
  BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
  [l],6*d-1)): od:

  for l from 0 to m - 2*3 do
  CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
  [l],6*d-1)): od:

  for l from 0 to m - 3*3 do
  DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
  [l],6*d-1)): od:

  for l from 0 to m - 4*3 do
  EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
  [l],6*d-1)): od;

  for l from 0 to m - 5*3 do
  FF||l := add(add(F||l[i,j]*x^i*y^j, i=0..dF[l]-j), j=0..min(dF
  [l],6*d-1)): od:

  for l from 0 to m - 6*3 do:
  GG||l := add(add(G||l[i,j]*x^i*y^j, i=0..dG[l]-j), j=0..min(dG
  [l],6*d-1)): od:

  for l from 0 to m - 7*3 do
  HH||l := add(add(H||l[i,j]*x^i*y^j, i=0..dH[l]-j), j=0..min(dH
  [l],6*d-1)): od:

  ## SYSTEM FOR DIVISIBILITY BY y^(6*d)

  Systemey6d := {}:

  for k from 0 to (m-6)/3   do

  Coeffs_y6d[6,m-6-3*k,k] := {coeffs(expand(mtaylor(Eq[6,m-6-3*k,
  k], y, 6*d)), [x,y])}:

  Systemey6d := Systemey6d union Coeffs_y6d[6,m-6-3*k,k]:

  od:

  Systeme_y6d := Systemey6d:

  ## VARIABLES FOR DIVISIBILITY BY y^(6*d)

  Variables_y6d := indets(Systeme_y6d):

  ## RESOLUTION y^(6*d)
```

```
Solutions_y6d := solve(Systeme_y6d, Variables_y6d):

assign(Solutions_y6d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;
for l from 0 to m - 5*3 do nops(expand(FF||l)) od;
for l from 0 to m - 6*3 do nops(expand(GG||l)) od;
for l from 0 to m - 7*3 do nops(expand(HH||l)) od;
```

$$l := 0$$

$$l := 0$$

$$407$$

$$l := 1$$

$$808$$

$$l := 2$$

$$1114$$

$$l := 3$$

$$1383$$

$$l := 4$$

$$1396$$

$$l := 5$$

$$1488$$

$$l := 6$$

$$1180$$

$$l := 7$$

$$3276$$

$$l := 8$$

$$3348$$

$$l := 9$$

$$l := 0$$

$$1$$

$$l := 1$$

$$1$$

$$l := 2$$

$$1$$

$$l := 3$$

$$1$$

$$l := 4$$

$$1$$

$$l := 5$$

$$1$$

$$l := 6$$

$$l := 0$$

$$1$$

$$l := 1$$

$$1$$

$$l := 2$$

$$1$$

$$l := 3$$

$$l := 0$$

$$l := 0$$

$$l := 0$$

$$l := 0$$

$$l := 0 \tag{7}$$

```
> ## EXPAND ALL UNKNOWNS UP TO y^(7*d)

  for l from 0 to m - 0*3 do
  AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
  [l],7*d-1)): od:

  for l from 0 to m - 1*3 do
  BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
  [l],7*d-1)): od:

  for l from 0 to m - 2*3 do
  CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
  [l],7*d-1)): od:

  for l from 0 to m - 3*3 do
  DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
  [l],7*d-1)): od:

  for l from 0 to m - 4*3 do
  EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
  [l],7*d-1)): od;

  for l from 0 to m - 5*3 do
  FF||l := add(add(F||l[i,j]*x^i*y^j, i=0..dF[l]-j), j=0..min(dF
  [l],7*d-1)): od:

  for l from 0 to m - 6*3 do:
```

```
GG||l := add(add(G||l[i,j]*x^i*y^j, i=0..dG[l]-j), j=0..min(dG
[l],7*d-1)): od:

for l from 0 to m - 7*3 do
HH||l := add(add(H||l[i,j]*x^i*y^j, i=0..dH[l]-j), j=0..min(dH
[l],7*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(7*d)

Systemey7d := {}:

for k from 0 to (m-7)/3   do

Coeffs_y7d[7,m-7-3*k,k] := {coeffs(expand(mtaylor(Eq[7,m-7-3*k,
k], y, 7*d)), [x,y])}:

Systemey7d := Systemey7d union Coeffs_y7d[7,m-7-3*k,k]:

od:

Systeme_y7d := Systemey7d:

## VARIABLES FOR DIVISIBILITY BY y^(7*d)

Variables_y7d := indets(Systeme_y7d):

## RESOLUTION y^(7*d)

Solutions_y7d := solve(Systeme_y7d, Variables_y7d):

assign(Solutions_y7d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;
for l from 0 to m - 5*3 do nops(expand(FF||l)) od;
for l from 0 to m - 6*3 do nops(expand(GG||l)) od;
for l from 0 to m - 7*3 do nops(expand(HH||l)) od;
```

$$l := 0$$

$$l := 0$$

$$1$$

$$l := 1$$

$$1$$

$$l := 2$$

$$1$$

$$l := 3$$

$$1$$

$$l := 4$$
$$1$$
$$l := 5$$
$$1$$
$$l := 6$$
$$1$$
$$l := 7$$
$$1$$
$$l := 8$$
$$3403$$
$$l := 9$$
$$l := 0$$
$$1$$
$$l := 1$$
$$1$$
$$l := 2$$
$$1$$
$$l := 3$$
$$1$$
$$l := 4$$
$$1$$
$$l := 5$$
$$1$$
$$l := 6$$
$$l := 0$$
$$1$$
$$l := 1$$
$$1$$
$$l := 2$$
$$1$$
$$l := 3$$
$$l := 0$$
$$l := 0$$
$$l := 0$$
$$l := 0$$
$$l := 0 \tag{8}$$

```
>
## EXPAND ALL UNKNOWNS UP TO y^(8*d)

for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
[l],8*d-1)): od:

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
[l],8*d-1)): od:

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
[l],8*d-1)): od:

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
[l],8*d-1)): od:

for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
[l],8*d-1)): od;

for l from 0 to m - 5*3 do
FF||l := add(add(F||l[i,j]*x^i*y^j, i=0..dF[l]-j), j=0..min(dF
[l],8*d-1)): od:

for l from 0 to m - 6*3 do:
GG||l := add(add(G||l[i,j]*x^i*y^j, i=0..dG[l]-j), j=0..min(dG
[l],8*d-1)): od:

for l from 0 to m - 7*3 do
HH||l := add(add(H||l[i,j]*x^i*y^j, i=0..dH[l]-j), j=0..min(dH
[l],8*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(8*d)

Systemey8d := {}:

for k from 0 to (m-8)/3   do

Coeffs_y8d[8,m-8-3*k,k] := {coeffs(expand(mtaylor(Eq[8,m-8-3*k,
k], y, 8*d)), [x,y])}:

Systemey8d := Systemey8d union Coeffs_y8d[8,m-8-3*k,k]:

od:

Systeme_y8d := Systemey8d:

## VARIABLES FOR DIVISIBILITY BY y^(8*d)

Variables_y8d := indets(Systeme_y8d):

## RESOLUTION y^(8*d)
```

```
Solutions_y8d := solve(Systeme_y8d, Variables_y8d):

assign(Solutions_y8d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;
for l from 0 to m - 5*3 do nops(expand(FF||l)) od;
for l from 0 to m - 6*3 do nops(expand(GG||l)) od;
for l from 0 to m - 7*3 do nops(expand(HH||l)) od;
```

$$l := 0$$

$$l := 0$$

$$1$$

$$l := 1$$

$$1$$

$$l := 2$$

$$1$$

$$l := 3$$

$$1$$

$$l := 4$$

$$1$$

$$l := 5$$

$$1$$

$$l := 6$$

$$1$$

$$l := 7$$

$$1$$

$$l := 8$$

$$1$$

$$l := 9$$

$$l := 0$$

$$1$$

$$l := 1$$

$$1$$

$$l := 2$$

$$1$$

$$l := 3$$

$$1$$

$$l := 4$$
$$1$$
$$l := 5$$
$$1$$
$$l := 6$$
$$l := 0$$
$$1$$
$$l := 1$$
$$1$$
$$l := 2$$
$$1$$
$$l := 3$$
$$l := 0$$
$$l := 0$$
$$l := 0$$
$$l := 0$$
$$l := 0 \tag{9}$$

```
> ## EXPAND ALL UNKNOWNS UP TO y^(9*d)

  for l from 0 to m - 0*3 do
  AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
  [l],9*d-1)): od:

  for l from 0 to m - 1*3 do
  BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
  [l],9*d-1)): od:

  for l from 0 to m - 2*3 do
  CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
  [l],9*d-1)): od:

  for l from 0 to m - 3*3 do
  DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
  [l],9*d-1)): od:

  for l from 0 to m - 4*3 do
  EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
  [l],9*d-1)): od;

  for l from 0 to m - 5*3 do
  FF||l := add(add(F||l[i,j]*x^i*y^j, i=0..dF[l]-j), j=0..min(dF
  [l],9*d-1)): od:

  for l from 0 to m - 6*3 do:
```

```
GG||l := add(add(G||l[i,j]*x^i*y^j, i=0..dG[l]-j), j=0..min(dG
[l],9*d-1)): od:

for l from 0 to m - 7*3 do
HH||l := add(add(H||l[i,j]*x^i*y^j, i=0..dH[l]-j), j=0..min(dH
[l],9*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(9*d)

Systemey9d := {}:

for k from 0 to (m-9)/3   do

Coeffs_y9d[9,m-9-3*k,k] := {coeffs(expand(mtaylor(Eq[9,m-9-3*k,
k], y, 9*d)), [x,y])}:

Systemey9d := Systemey9d union Coeffs_y9d[9,m-9-3*k,k]:

od:

Systeme_y9d := Systemey9d:

## VARIABLES FOR DIVISIBILITY BY y^(9*d)

Variables_y9d := indets(Systeme_y9d):

## RESOLUTION y^(9*d)

Solutions_y9d := solve(Systeme_y9d, Variables_y9d):

assign(Solutions_y9d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;
for l from 0 to m - 5*3 do nops(expand(FF||l)) od;
for l from 0 to m - 6*3 do nops(expand(GG||l)) od;
for l from 0 to m - 7*3 do nops(expand(HH||l)) od;
```

$$l := 0$$

$$l := 0$$

$$1$$

$$l := 1$$

$$1$$

$$l := 2$$

$$1$$

$$l := 3$$

$$1$$

$$
\begin{aligned}
&l := 4\\
&\quad 1\\
&l := 5\\
&\quad 1\\
&l := 6\\
&\quad 1\\
&l := 7\\
&\quad 1\\
&l := 8\\
&\quad 1\\
&l := 9\\
&l := 0\\
&\quad 1\\
&l := 1\\
&\quad 1\\
&l := 2\\
&\quad 1\\
&l := 3\\
&\quad 1\\
&l := 4\\
&\quad 1\\
&l := 5\\
&\quad 1\\
&l := 6\\
&l := 0\\
&\quad 1\\
&l := 1\\
&\quad 1\\
&l := 2\\
&\quad 1\\
&l := 3\\
&l := 0\\
&l := 0\\
&l := 0\\
&l := 0\\
&l := 0
\end{aligned}
\tag{10}
$$

```
>
## EXPAND ALL UNKNOWNS UP TO y^(10*d)

for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
[l],10*d-1)): od:

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
[l],10*d-1)): od:

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
[l],10*d-1)): od:

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
[l],10*d-1)): od:

for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
[l],10*d-1)): od;

for l from 0 to m - 5*3 do
FF||l := add(add(F||l[i,j]*x^i*y^j, i=0..dF[l]-j), j=0..min(dF
[l],10*d-1)): od:

for l from 0 to m - 6*3 do:
GG||l := add(add(G||l[i,j]*x^i*y^j, i=0..dG[l]-j), j=0..min(dG
[l],10*d-1)): od:

for l from 0 to m - 7*3 do
HH||l := add(add(H||l[i,j]*x^i*y^j, i=0..dH[l]-j), j=0..min(dH
[l],10*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(10*d)

Systemey10d := {}:

for k from 0 to (m-10)/3   do

Coeffs_y10d[10,m-10-3*k,k] := {coeffs(expand(mtaylor(Eq[10,m-10
-3*k,k], y, 10*d)), [x,y])}:

Systemey10d := Systemey10d union Coeffs_y10d[10,m-10-3*k,k]:

od:

Systeme_y10d := Systemey10d:

## VARIABLES FOR DIVISIBILITY BY y^(10*d)

Variables_y10d := indets(Systeme_y10d):

## RESOLUTION y^(10*d)
```

```
Solutions_y10d := solve(Systeme_y10d, Variables_y10d):

assign(Solutions_y10d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;
for l from 0 to m - 5*3 do nops(expand(FF||l)) od;
for l from 0 to m - 6*3 do nops(expand(GG||l)) od;
for l from 0 to m - 7*3 do nops(expand(HH||l)) od;
```

$$l := 0$$

$$l := 0$$

$$1$$

$$l := 1$$

$$1$$

$$l := 2$$

$$1$$

$$l := 3$$

$$1$$

$$l := 4$$

$$1$$

$$l := 5$$

$$1$$

$$l := 6$$

$$1$$

$$l := 7$$

$$1$$

$$l := 8$$

$$1$$

$$l := 9$$

$$l := 0$$

$$1$$

$$l := 1$$

$$1$$

$$l := 2$$

$$1$$

$$l := 3$$

$$1$$

$$l := 4$$

$$1$$

$$l := 5$$

$$1$$

$$l := 6$$

$$l := 0$$

$$1$$

$$l := 1$$

$$1$$

$$l := 2$$

$$1$$

$$l := 3$$

$$l := 0$$

$$l := 0$$

$$l := 0$$

$$l := 0$$

$$l := 0 \tag{11}$$

```
> ## EXPAND ALL UNKNOWNS UP TO y^(11*d)

  for l from 0 to m - 0*3 do
  AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
  [l],11*d-1)): od:

  for l from 0 to m - 1*3 do
  BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
  [l],11*d-1)): od:

  for l from 0 to m - 2*3 do
  CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
  [l],11*d-1)): od:

  for l from 0 to m - 3*3 do
  DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
  [l],11*d-1)): od:

  for l from 0 to m - 4*3 do
  EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
  [l],11*d-1)): od;

  for l from 0 to m - 5*3 do
  FF||l := add(add(F||l[i,j]*x^i*y^j, i=0..dF[l]-j), j=0..min(dF
  [l],11*d-1)): od:
```

```
for l from 0 to m - 6*3 do:
GG||l := add(add(G||l[i,j]*x^i*y^j, i=0..dG[l]-j), j=0..min(dG
[l],11*d-1)): od:

for l from 0 to m - 7*3 do
HH||l := add(add(H||l[i,j]*x^i*y^j, i=0..dH[l]-j), j=0..min(dH
[l],11*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(11*d)

Systemey11d := {}:

for k from 0 to (m-11)/3   do

Coeffs_y11d[11,m-11-3*k,k] := {coeffs(expand(mtaylor(Eq[11,m-11
-3*k,k], y, 11*d)), [x,y])}:

Systemey11d := Systemey11d union Coeffs_y11d[11,m-11-3*k,k]:

od:

Systeme_y11d := Systemey11d:

## VARIABLES FOR DIVISIBILITY BY y^(11*d)

Variables_y11d := indets(Systeme_y11d):

## RESOLUTION y^(11*d)

Solutions_y11d := solve(Systeme_y11d, Variables_y11d):

assign(Solutions_y11d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;
for l from 0 to m - 5*3 do nops(expand(FF||l)) od;
for l from 0 to m - 6*3 do nops(expand(GG||l)) od;
for l from 0 to m - 7*3 do nops(expand(HH||l)) od;
```

$$l := 0$$

$$l := 0$$

$$1$$

$$l := 1$$

$$1$$

$$l := 2$$

$$1$$

$$l := 3$$

$$
\begin{aligned}
&1\\
&l := 4\\
&1\\
&l := 5\\
&1\\
&l := 6\\
&1\\
&l := 7\\
&1\\
&l := 8\\
&1\\
&l := 9\\
&l := 0\\
&1\\
&l := 1\\
&1\\
&l := 2\\
&1\\
&l := 3\\
&1\\
&l := 4\\
&1\\
&l := 5\\
&1\\
&l := 6\\
&l := 0\\
&1\\
&l := 1\\
&1\\
&l := 2\\
&1\\
&l := 3\\
&l := 0\\
&l := 0\\
&l := 0\\
&l := 0\\
&l := 0
\end{aligned}
\tag{12}
$$

```
> ## EXPAND ALL UNKNOWNS UP TO y^(12*d)

  for l from 0 to m - 0*3 do
  AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
  [l],12*d-1)): od:

  for l from 0 to m - 1*3 do
  BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
  [l],12*d-1)): od:

  for l from 0 to m - 2*3 do
  CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
  [l],12*d-1)): od:

  for l from 0 to m - 3*3 do
  DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
  [l],12*d-1)): od:

  for l from 0 to m - 4*3 do
  EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
  [l],12*d-1)): od;

  for l from 0 to m - 5*3 do
  FF||l := add(add(F||l[i,j]*x^i*y^j, i=0..dF[l]-j), j=0..min(dF
  [l],12*d-1)): od:

  for l from 0 to m - 6*3 do:
  GG||l := add(add(G||l[i,j]*x^i*y^j, i=0..dG[l]-j), j=0..min(dG
  [l],12*d-1)): od:

  for l from 0 to m - 7*3 do
  HH||l := add(add(H||l[i,j]*x^i*y^j, i=0..dH[l]-j), j=0..min(dH
  [l],12*d-1)): od:

  ## SYSTEM FOR DIVISIBILITY BY y^(12*d)

  Systemey12d := {}:

  for k from 0 to (m-12)/3   do

  Coeffs_y12d[12,m-12-3*k,k] := {coeffs(expand(mtaylor(Eq[12,m-12
  -3*k,k], y, 12*d)), [x,y])}:

  Systemey12d := Systemey12d union Coeffs_y12d[12,m-12-3*k,k]:

  od:

  Systeme_y12d := Systemey12d:

  ## VARIABLES FOR DIVISIBILITY BY y^(12*d)

  Variables_y12d := indets(Systeme_y12d):

  ## RESOLUTION y^(12*d)
```

```
Solutions_y12d := solve(Systeme_y12d, Variables_y12d):

assign(Solutions_y12d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;
for l from 0 to m - 5*3 do nops(expand(FF||l)) od;
for l from 0 to m - 6*3 do nops(expand(GG||l)) od;
for l from 0 to m - 7*3 do nops(expand(HH||l)) od;
```

$$l := 0$$

$$l := 0$$

$$1$$

$$l := 1$$

$$1$$

$$l := 2$$

$$1$$

$$l := 3$$

$$1$$

$$l := 4$$

$$1$$

$$l := 5$$

$$1$$

$$l := 6$$

$$1$$

$$l := 7$$

$$1$$

$$l := 8$$

$$1$$

$$l := 9$$

$$l := 0$$

$$1$$

$$l := 1$$

$$1$$

$$l := 2$$

$$1$$

$$l := 3$$

$$1$$

$$l := 4$$
$$1$$
$$l := 5$$
$$1$$
$$l := 6$$
$$l := 0$$
$$1$$
$$l := 1$$
$$1$$
$$l := 2$$
$$1$$
$$l := 3$$
$$l := 0$$
$$l := 0$$
$$l := 0$$
$$l := 0$$
$$l := 0 \tag{13}$$

```
> ## EXPAND ALL UNKNOWNS UP TO y^(13*d)

  for l from 0 to m - 0*3 do
  AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
  [l],13*d-1)): od:

  for l from 0 to m - 1*3 do
  BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
  [l],13*d-1)): od:

  for l from 0 to m - 2*3 do
  CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
  [l],13*d-1)): od:

  for l from 0 to m - 3*3 do
  DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
  [l],13*d-1)): od:

  for l from 0 to m - 4*3 do
  EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
  [l],13*d-1)): od;

  for l from 0 to m - 5*3 do
  FF||l := add(add(F||l[i,j]*x^i*y^j, i=0..dF[l]-j), j=0..min(dF
  [l],13*d-1)): od:

  for l from 0 to m - 6*3 do:
```

```
GG||l := add(add(G||l[i,j]*x^i*y^j, i=0..dG[l]-j), j=0..min(dG
[l],13*d-1)): od:

for l from 0 to m - 7*3 do
HH||l := add(add(H||l[i,j]*x^i*y^j, i=0..dH[l]-j), j=0..min(dH
[l],13*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(13*d)

Systemey13d := {}:

for k from 0 to (m-13)/3   do

Coeffs_y13d[13,m-13-3*k,k] := {coeffs(expand(mtaylor(Eq[13,m-13
-3*k,k], y, 13*d)), [x,y])}:

Systemey13d := Systemey13d union Coeffs_y13d[13,m-13-3*k,k]:

od:

Systeme_y13d := Systemey13d:

## VARIABLES FOR DIVISIBILITY BY y^(13*d)

Variables_y13d := indets(Systeme_y13d):

## RESOLUTION y^(13*d)

Solutions_y13d := solve(Systeme_y13d, Variables_y13d):

assign(Solutions_y13d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;
for l from 0 to m - 5*3 do nops(expand(FF||l)) od;
for l from 0 to m - 6*3 do nops(expand(GG||l)) od;
for l from 0 to m - 7*3 do nops(expand(HH||l)) od;
```

$$l := 0$$

$$l := 0$$

$$1$$

$$l := 1$$

$$1$$

$$l := 2$$

$$1$$

$$l := 3$$

$$1$$

$$
\begin{aligned}
&l := 4\\
&\quad 1\\
&l := 5\\
&\quad 1\\
&l := 6\\
&\quad 1\\
&l := 7\\
&\quad 1\\
&l := 8\\
&\quad 1\\
&l := 9\\
&l := 0\\
&\quad 1\\
&l := 1\\
&\quad 1\\
&l := 2\\
&\quad 1\\
&l := 3\\
&\quad 1\\
&l := 4\\
&\quad 1\\
&l := 5\\
&\quad 1\\
&l := 6\\
&l := 0\\
&\quad 1\\
&l := 1\\
&\quad 1\\
&l := 2\\
&\quad 1\\
&l := 3\\
&l := 0\\
&l := 0\\
&l := 0\\
&l := 0\\
&l := 0
\end{aligned}
\tag{14}
$$

```
> ## EXPAND ALL UNKNOWNS UP TO y^(14*d)

  for l from 0 to m - 0*3 do
  AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
  [l],14*d-1)): od:

  for l from 0 to m - 1*3 do
  BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
  [l],14*d-1)): od:

  for l from 0 to m - 2*3 do
  CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
  [l],14*d-1)): od:

  for l from 0 to m - 3*3 do
  DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
  [l],14*d-1)): od:

  for l from 0 to m - 4*3 do
  EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
  [l],14*d-1)): od;

  for l from 0 to m - 5*3 do
  FF||l := add(add(F||l[i,j]*x^i*y^j, i=0..dF[l]-j), j=0..min(dF
  [l],14*d-1)): od:

  for l from 0 to m - 6*3 do:
  GG||l := add(add(G||l[i,j]*x^i*y^j, i=0..dG[l]-j), j=0..min(dG
  [l],14*d-1)): od:

  for l from 0 to m - 7*3 do
  HH||l := add(add(H||l[i,j]*x^i*y^j, i=0..dH[l]-j), j=0..min(dH
  [l],14*d-1)): od:

  ## SYSTEM FOR DIVISIBILITY BY y^(14*d)

  Systemey14d := {}:

  for k from 0 to (m-14)/3   do

  Coeffs_y14d[14,m-14-3*k,k] := {coeffs(expand(mtaylor(Eq[14,m-14
  -3*k,k], y, 14*d)), [x,y])}:

  Systemey14d := Systemey14d union Coeffs_y14d[14,m-14-3*k,k]:

  od:

  Systeme_y14d := Systemey14d:

  ## VARIABLES FOR DIVISIBILITY BY y^(14*d)

  Variables_y14d := indets(Systeme_y14d):

  ## RESOLUTION y^(14*d)

  Solutions_y14d := solve(Systeme_y14d, Variables_y14d):
```

```
assign(Solutions_y14d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;
for l from 0 to m - 5*3 do nops(expand(FF||l)) od;
for l from 0 to m - 6*3 do nops(expand(GG||l)) od;
for l from 0 to m - 7*3 do nops(expand(HH||l)) od;
```

$$l := 0$$

$$l := 0$$

$$1$$

$$l := 1$$

$$1$$

$$l := 2$$

$$1$$

$$l := 3$$

$$1$$

$$l := 4$$

$$1$$

$$l := 5$$

$$1$$

$$l := 6$$

$$1$$

$$l := 7$$

$$1$$

$$l := 8$$

$$1$$

$$l := 9$$

$$l := 0$$

$$1$$

$$l := 1$$

$$1$$

$$l := 2$$

$$1$$

$$l := 3$$

$$1$$

$$l := 4$$

$$1$$
$$l := 5$$
$$1$$
$$l := 6$$
$$l := 0$$
$$1$$
$$l := 1$$
$$1$$
$$l := 2$$
$$1$$
$$l := 3$$
$$l := 0$$
$$l := 0$$
$$l := 0$$
$$l := 0$$
$$l := 0 \tag{15}$$

```
> ## EXPAND ALL UNKNOWNS UP TO y^(15*d)

  for l from 0 to m - 0*3 do
  AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
  [l],15*d-1)): od:

  for l from 0 to m - 1*3 do
  BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
  [l],15*d-1)): od:

  for l from 0 to m - 2*3 do
  CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
  [l],15*d-1)): od:

  for l from 0 to m - 3*3 do
  DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
  [l],15*d-1)): od:

  for l from 0 to m - 4*3 do
  EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
  [l],15*d-1)): od;

  for l from 0 to m - 5*3 do
  FF||l := add(add(F||l[i,j]*x^i*y^j, i=0..dF[l]-j), j=0..min(dF
  [l],15*d-1)): od:

  for l from 0 to m - 6*3 do:
  GG||l := add(add(G||l[i,j]*x^i*y^j, i=0..dG[l]-j), j=0..min(dG
  [l],15*d-1)): od:
```

```
for l from 0 to m - 7*3 do
HH||l := add(add(H||l[i,j]*x^i*y^j, i=0..dH[l]-j), j=0..min(dH
[l],15*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(15*d)

Systemey15d := {}:

for k from 0 to (m-15)/3   do

Coeffs_y15d[15,m-15-3*k,k] := {coeffs(expand(mtaylor(Eq[15,m-15
-3*k,k], y, 15*d)), [x,y])}:

Systemey15d := Systemey15d union Coeffs_y15d[15,m-15-3*k,k]:

od:

Systeme_y15d := Systemey15d:

## VARIABLES FOR DIVISIBILITY BY y^(15*d)

Variables_y15d := indets(Systeme_y15d):

## RESOLUTION y^(15*d)

Solutions_y15d := solve(Systeme_y15d, Variables_y15d):

assign(Solutions_y15d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;
for l from 0 to m - 5*3 do nops(expand(FF||l)) od;
for l from 0 to m - 6*3 do nops(expand(GG||l)) od;
for l from 0 to m - 7*3 do nops(expand(HH||l)) od;
```

$$l := 0$$

$$l := 0$$

$$1$$

$$l := 1$$

$$1$$

$$l := 2$$

$$1$$

$$l := 3$$

$$1$$

$$l := 4$$

$$1$$

$$l := 5$$
$$1$$
$$l := 6$$
$$1$$
$$l := 7$$
$$1$$
$$l := 8$$
$$1$$
$$l := 9$$
$$l := 0$$
$$1$$
$$l := 1$$
$$1$$
$$l := 2$$
$$1$$
$$l := 3$$
$$1$$
$$l := 4$$
$$1$$
$$l := 5$$
$$1$$
$$l := 6$$
$$l := 0$$
$$1$$
$$l := 1$$
$$1$$
$$l := 2$$
$$1$$
$$l := 3$$
$$l := 0$$
$$l := 0$$
$$l := 0$$
$$l := 0$$
$$l := 0 \tag{16}$$

```
> ## EXPAND ALL UNKNOWNS UP TO y^(16*d)
```

```
for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
[l],16*d-1)): od:

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
[l],16*d-1)): od:

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
[l],16*d-1)): od:

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
[l],16*d-1)): od:

for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
[l],16*d-1)): od;

for l from 0 to m - 5*3 do
FF||l := add(add(F||l[i,j]*x^i*y^j, i=0..dF[l]-j), j=0..min(dF
[l],16*d-1)): od:

for l from 0 to m - 6*3 do:
GG||l := add(add(G||l[i,j]*x^i*y^j, i=0..dG[l]-j), j=0..min(dG
[l],16*d-1)): od:

for l from 0 to m - 7*3 do
HH||l := add(add(H||l[i,j]*x^i*y^j, i=0..dH[l]-j), j=0..min(dH
[l],16*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(16*d)

Systemey16d := {}:

for k from 0 to (m-16)/3   do

Coeffs_y16d[16,m-16-3*k,k] := {coeffs(expand(mtaylor(Eq[16,m-16
-3*k,k], y, 16*d)), [x,y])}:

Systemey16d := Systemey16d union Coeffs_y16d[16,m-16-3*k,k]:

od:

Systeme_y16d := Systemey16d:

## VARIABLES FOR DIVISIBILITY BY y^(16*d)

Variables_y16d := indets(Systeme_y16d):

## RESOLUTION y^(16*d)

Solutions_y16d := solve(Systeme_y16d, Variables_y16d):

assign(Solutions_y16d):
```

```
##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;
for l from 0 to m - 5*3 do nops(expand(FF||l)) od;
for l from 0 to m - 6*3 do nops(expand(GG||l)) od;
for l from 0 to m - 7*3 do nops(expand(HH||l)) od;
```

$$l := 0$$
$$l := 0$$
$$1$$
$$l := 1$$
$$1$$
$$l := 2$$
$$1$$
$$l := 3$$
$$1$$
$$l := 4$$
$$1$$
$$l := 5$$
$$1$$
$$l := 6$$
$$1$$
$$l := 7$$
$$1$$
$$l := 8$$
$$1$$
$$l := 9$$
$$l := 0$$
$$1$$
$$l := 1$$
$$1$$
$$l := 2$$
$$1$$
$$l := 3$$
$$1$$
$$l := 4$$
$$1$$

$$l := 5$$

$$1$$

$$l := 6$$

$$l := 0$$

$$1$$

$$l := 1$$

$$1$$

$$l := 2$$

$$1$$

$$l := 3$$

$$l := 0$$

$$l := 0$$

$$l := 0$$

$$l := 0$$

$$l := 0 \tag{17}$$

```
> ## EXPAND ALL UNKNOWNS UP TO y^(17*d)

  for l from 0 to m - 0*3 do
  AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
  [l],17*d-1)): od:

  for l from 0 to m - 1*3 do
  BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
  [l],17*d-1)): od:

  for l from 0 to m - 2*3 do
  CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
  [l],17*d-1)): od:

  for l from 0 to m - 3*3 do
  DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
  [l],17*d-1)): od:

  for l from 0 to m - 4*3 do
  EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
  [l],17*d-1)): od;

  for l from 0 to m - 5*3 do
  FF||l := add(add(F||l[i,j]*x^i*y^j, i=0..dF[l]-j), j=0..min(dF
  [l],17*d-1)): od:

  for l from 0 to m - 6*3 do:
  GG||l := add(add(G||l[i,j]*x^i*y^j, i=0..dG[l]-j), j=0..min(dG
  [l],17*d-1)): od:
```

```
for l from 0 to m - 7*3 do
HH||l := add(add(H||l[i,j]*x^i*y^j, i=0..dH[l]-j), j=0..min(dH
[l],17*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(17*d)

Systemey17d := {}:

for k from 0 to (m-17)/3   do

Coeffs_y17d[17,m-17-3*k,k] := {coeffs(expand(mtaylor(Eq[17,m-17
-3*k,k], y, 17*d)), [x,y])}:

Systemey17d := Systemey17d union Coeffs_y17d[17,m-17-3*k,k]:

od:

Systeme_y17d := Systemey17d:

## VARIABLES FOR DIVISIBILITY BY y^(17*d)

Variables_y17d := indets(Systeme_y17d):

## RESOLUTION y^(17*d)

Solutions_y17d := solve(Systeme_y17d, Variables_y17d):

assign(Solutions_y17d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;
for l from 0 to m - 5*3 do nops(expand(FF||l)) od;
for l from 0 to m - 6*3 do nops(expand(GG||l)) od;
for l from 0 to m - 7*3 do nops(expand(HH||l)) od;
```

$$l := 0$$

$$l := 0$$

$$1$$

$$l := 1$$

$$1$$

$$l := 2$$

$$1$$

$$l := 3$$

$$1$$

$$l := 4$$

$$1$$

$$l := 5$$
$$1$$
$$l := 6$$
$$1$$
$$l := 7$$
$$1$$
$$l := 8$$
$$1$$
$$l := 9$$
$$l := 0$$
$$1$$
$$l := 1$$
$$1$$
$$l := 2$$
$$1$$
$$l := 3$$
$$1$$
$$l := 4$$
$$1$$
$$l := 5$$
$$1$$
$$l := 6$$
$$l := 0$$
$$1$$
$$l := 1$$
$$1$$
$$l := 2$$
$$1$$
$$l := 3$$
$$l := 0$$
$$l := 0$$
$$l := 0$$
$$l := 0$$
$$l := 0 \tag{18}$$

```
> ## EXPAND ALL UNKNOWNS UP TO y^(18*d)
```

```
for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
[l],18*d-1)): od:

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
[l],18*d-1)): od:

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
[l],18*d-1)): od:

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
[l],18*d-1)): od:

for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
[l],18*d-1)): od;

for l from 0 to m - 5*3 do
FF||l := add(add(F||l[i,j]*x^i*y^j, i=0..dF[l]-j), j=0..min(dF
[l],18*d-1)): od:

for l from 0 to m - 6*3 do:
GG||l := add(add(G||l[i,j]*x^i*y^j, i=0..dG[l]-j), j=0..min(dG
[l],18*d-1)): od:

for l from 0 to m - 7*3 do
HH||l := add(add(H||l[i,j]*x^i*y^j, i=0..dH[l]-j), j=0..min(dH
[l],18*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(18*d)

Systemey18d := {}:

for k from 0 to (m-18)/3   do

Coeffs_y18d[18,m-18-3*k,k] := {coeffs(expand(mtaylor(Eq[18,m-18
-3*k,k], y, 18*d)), [x,y])}:

Systemey18d := Systemey18d union Coeffs_y18d[18,m-18-3*k,k]:

od:

Systeme_y18d := Systemey18d:

## VARIABLES FOR DIVISIBILITY BY y^(18*d)

Variables_y18d := indets(Systeme_y18d):

## RESOLUTION y^(18*d)

Solutions_y18d := solve(Systeme_y18d, Variables_y18d):

assign(Solutions_y18d):
```

```
##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;
for l from 0 to m - 5*3 do nops(expand(FF||l)) od;
for l from 0 to m - 6*3 do nops(expand(GG||l)) od;
for l from 0 to m - 7*3 do nops(expand(HH||l)) od;
```

$$l := 0$$
$$l := 0$$
$$1$$
$$l := 1$$
$$1$$
$$l := 2$$
$$1$$
$$l := 3$$
$$1$$
$$l := 4$$
$$1$$
$$l := 5$$
$$1$$
$$l := 6$$
$$1$$
$$l := 7$$
$$1$$
$$l := 8$$
$$1$$
$$l := 9$$
$$l := 0$$
$$1$$
$$l := 1$$
$$1$$
$$l := 2$$
$$1$$
$$l := 3$$
$$1$$
$$l := 4$$
$$1$$

$$l := 5$$

$$1$$

$$l := 6$$

$$l := 0$$

$$1$$

$$l := 1$$

$$1$$

$$l := 2$$

$$1$$

$$l := 3$$

$$l := 0$$

$$l := 0$$

$$l := 0$$

$$l := 0$$

$$l := 0 \tag{19}$$

```
> ## EXPAND ALL UNKNOWNS UP TO y^(19*d)

  for l from 0 to m - 0*3 do
  AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
  [l],19*d-1)): od:

  for l from 0 to m - 1*3 do
  BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
  [l],19*d-1)): od:

  for l from 0 to m - 2*3 do
  CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
  [l],19*d-1)): od:

  for l from 0 to m - 3*3 do
  DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
  [l],19*d-1)): od:

  for l from 0 to m - 4*3 do
  EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
  [l],19*d-1)): od;

  for l from 0 to m - 5*3 do
  FF||l := add(add(F||l[i,j]*x^i*y^j, i=0..dF[l]-j), j=0..min(dF
  [l],19*d-1)): od:

  for l from 0 to m - 6*3 do:
  GG||l := add(add(G||l[i,j]*x^i*y^j, i=0..dG[l]-j), j=0..min(dG
  [l],19*d-1)): od:
```

```
for l from 0 to m - 7*3 do
HH||l := add(add(H||l[i,j]*x^i*y^j, i=0..dH[l]-j), j=0..min(dH
[l],19*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(19*d)

Systemey19d := {}:

for k from 0 to (m-19)/3   do

Coeffs_y19d[19,m-19-3*k,k] := {coeffs(expand(mtaylor(Eq[19,m-19
-3*k,k], y, 19*d)), [x,y])}:

Systemey19d := Systemey19d union Coeffs_y19d[19,m-19-3*k,k]:

od:

Systeme_y19d := Systemey19d:

## VARIABLES FOR DIVISIBILITY BY y^(19*d)

Variables_y19d := indets(Systeme_y19d):

## RESOLUTION y^(19*d)

Solutions_y19d := solve(Systeme_y19d, Variables_y19d):

assign(Solutions_y19d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;
for l from 0 to m - 5*3 do nops(expand(FF||l)) od;
for l from 0 to m - 6*3 do nops(expand(GG||l)) od;
for l from 0 to m - 7*3 do nops(expand(HH||l)) od;
```

$$l := 0$$

$$l := 0$$

$$1$$

$$l := 1$$

$$1$$

$$l := 2$$

$$1$$

$$l := 3$$

$$1$$

$$l := 4$$

$$1$$

$$l := 5$$
$$1$$
$$l := 6$$
$$1$$
$$l := 7$$
$$1$$
$$l := 8$$
$$1$$
$$l := 9$$
$$l := 0$$
$$1$$
$$l := 1$$
$$1$$
$$l := 2$$
$$1$$
$$l := 3$$
$$1$$
$$l := 4$$
$$1$$
$$l := 5$$
$$1$$
$$l := 6$$
$$l := 0$$
$$1$$
$$l := 1$$
$$1$$
$$l := 2$$
$$1$$
$$l := 3$$
$$l := 0$$
$$l := 0$$
$$l := 0$$
$$l := 0$$
$$l := 0 \tag{20}$$

```
> ## EXPAND ALL UNKNOWNS UP TO y^(20*d)
```

```
for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
[l],20*d-1)): od:

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
[l],20*d-1)): od:

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
[l],20*d-1)): od:

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
[l],20*d-1)): od:

for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
[l],20*d-1)): od;

for l from 0 to m - 5*3 do
FF||l := add(add(F||l[i,j]*x^i*y^j, i=0..dF[l]-j), j=0..min(dF
[l],20*d-1)): od:

for l from 0 to m - 6*3 do:
GG||l := add(add(G||l[i,j]*x^i*y^j, i=0..dG[l]-j), j=0..min(dG
[l],20*d-1)): od:

for l from 0 to m - 7*3 do
HH||l := add(add(H||l[i,j]*x^i*y^j, i=0..dH[l]-j), j=0..min(dH
[l],20*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(20*d)

Systemey20d := {}:

for k from 0 to (m-20)/3   do

Coeffs_y20d[20,m-20-3*k,k] := {coeffs(expand(mtaylor(Eq[20,m-20
-3*k,k], y, 20*d)), [x,y])}:

Systemey20d := Systemey20d union Coeffs_y20d[20,m-20-3*k,k]:

od:

Systeme_y20d := Systemey20d:

## VARIABLES FOR DIVISIBILITY BY y^(20*d)

Variables_y20d := indets(Systeme_y20d):

## RESOLUTION y^(20*d)

Solutions_y20d := solve(Systeme_y20d, Variables_y20d):

assign(Solutions_y20d):
```

```
##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;
for l from 0 to m - 5*3 do nops(expand(FF||l)) od;
for l from 0 to m - 6*3 do nops(expand(GG||l)) od;
for l from 0 to m - 7*3 do nops(expand(HH||l)) od;
```

$$l := 0$$
$$l := 0$$
$$1$$
$$l := 1$$
$$1$$
$$l := 2$$
$$1$$
$$l := 3$$
$$1$$
$$l := 4$$
$$1$$
$$l := 5$$
$$1$$
$$l := 6$$
$$1$$
$$l := 7$$
$$1$$
$$l := 8$$
$$1$$
$$l := 9$$
$$l := 0$$
$$1$$
$$l := 1$$
$$1$$
$$l := 2$$
$$1$$
$$l := 3$$
$$1$$
$$l := 4$$
$$1$$

$$l := 5$$

$$1$$

$$l := 6$$

$$l := 0$$

$$1$$

$$l := 1$$

$$1$$

$$l := 2$$

$$1$$

$$l := 3$$

$$l := 0$$

$$l := 0$$

$$l := 0$$

$$l := 0$$

$$l := 0 \tag{21}$$

```
> ## EXPAND ALL UNKNOWNS UP TO y^(21*d)

  for l from 0 to m - 0*3 do
  AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
  [l],21*d-1)): od:

  for l from 0 to m - 1*3 do
  BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
  [l],21*d-1)): od:

  for l from 0 to m - 2*3 do
  CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
  [l],21*d-1)): od:

  for l from 0 to m - 3*3 do
  DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
  [l],21*d-1)): od:

  for l from 0 to m - 4*3 do
  EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
  [l],21*d-1)): od;

  for l from 0 to m - 5*3 do
  FF||l := add(add(F||l[i,j]*x^i*y^j, i=0..dF[l]-j), j=0..min(dF
  [l],21*d-1)): od:

  for l from 0 to m - 6*3 do:
  GG||l := add(add(G||l[i,j]*x^i*y^j, i=0..dG[l]-j), j=0..min(dG
  [l],21*d-1)): od:
```

```
for l from 0 to m - 7*3 do
HH||l := add(add(H||l[i,j]*x^i*y^j, i=0..dH[l]-j), j=0..min(dH
[l],21*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(21*d)

Systemey21d := {}:

for k from 0 to (m-21)/3   do

Coeffs_y21d[21,m-21-3*k,k] := {coeffs(expand(mtaylor(Eq[21,m-21
-3*k,k], y, 21*d)), [x,y])}:

Systemey21d := Systemey21d union Coeffs_y21d[21,m-21-3*k,k]:

od:

Systeme_y21d := Systemey21d:

## VARIABLES FOR DIVISIBILITY BY y^(21*d)

Variables_y21d := indets(Systeme_y21d):

## RESOLUTION y^(21*d)

Solutions_y21d := solve(Systeme_y21d, Variables_y21d):

assign(Solutions_y21d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;
for l from 0 to m - 5*3 do nops(expand(FF||l)) od;
for l from 0 to m - 6*3 do nops(expand(GG||l)) od;
for l from 0 to m - 7*3 do nops(expand(HH||l)) od;
```

$$l := 0$$

$$l := 0$$

$$1$$

$$l := 1$$

$$1$$

$$l := 2$$

$$1$$

$$l := 3$$

$$1$$

$$l := 4$$

$$
\begin{aligned}
&1\\
&l := 5\\
&1\\
&l := 6\\
&1\\
&l := 7\\
&1\\
&l := 8\\
&1\\
&l := 9\\
&l := 0\\
&1\\
&l := 1\\
&1\\
&l := 2\\
&1\\
&l := 3\\
&1\\
&l := 4\\
&1\\
&l := 5\\
&1\\
&l := 6\\
&l := 0\\
&1\\
&l := 1\\
&1\\
&l := 2\\
&1\\
&l := 3\\
&l := 0\\
&l := 0\\
&l := 0\\
&l := 0\\
&l := 0
\end{aligned}
\tag{22}
$$

```
## EXPAND ALL UNKNOWNS UP TO y^(22*d)

for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
[l],22*d-1)): od:

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
[l],22*d-1)): od:

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
[l],22*d-1)): od:

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
[l],22*d-1)): od:

for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
[l],22*d-1)): od;

for l from 0 to m - 5*3 do
FF||l := add(add(F||l[i,j]*x^i*y^j, i=0..dF[l]-j), j=0..min(dF
[l],22*d-1)): od:

for l from 0 to m - 6*3 do:
GG||l := add(add(G||l[i,j]*x^i*y^j, i=0..dG[l]-j), j=0..min(dG
[l],22*d-1)): od:

for l from 0 to m - 7*3 do
HH||l := add(add(H||l[i,j]*x^i*y^j, i=0..dH[l]-j), j=0..min(dH
[l],22*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(22*d)

Systemey22d := {}:

for k from 0 to (m-22)/3   do

Coeffs_y22d[22,m-22-3*k,k] := {coeffs(expand(mtaylor(Eq[22,m-22
-3*k,k], y, 22*d)), [x,y])}:

Systemey22d := Systemey22d union Coeffs_y22d[22,m-22-3*k,k]:

od:

Systeme_y22d := Systemey22d:

## VARIABLES FOR DIVISIBILITY BY y^(22*d)

Variables_y22d := indets(Systeme_y22d):

## RESOLUTION y^(22*d)

Solutions_y22d := solve(Systeme_y22d, Variables_y22d):
```

```
assign(Solutions_y22d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;
for l from 0 to m - 5*3 do nops(expand(FF||l)) od;
for l from 0 to m - 6*3 do nops(expand(GG||l)) od;
for l from 0 to m - 7*3 do nops(expand(HH||l)) od;
```

$$l := 0$$
$$l := 0$$
$$1$$
$$l := 1$$
$$1$$
$$l := 2$$
$$1$$
$$l := 3$$
$$1$$
$$l := 4$$
$$1$$
$$l := 5$$
$$1$$
$$l := 6$$
$$1$$
$$l := 7$$
$$1$$
$$l := 8$$
$$1$$
$$l := 9$$
$$l := 0$$
$$1$$
$$l := 1$$
$$1$$
$$l := 2$$
$$1$$
$$l := 3$$
$$1$$
$$l := 4$$

$$
\begin{aligned}
&1\\
&l := 5\\
&1\\
&l := 6\\
&l := 0\\
&1\\
&l := 1\\
&1\\
&l := 2\\
&1\\
&l := 3\\
&l := 0\\
&l := 0\\
&l := 0\\
&l := 0\\
&l := 0
\end{aligned}
\tag{23}
$$